

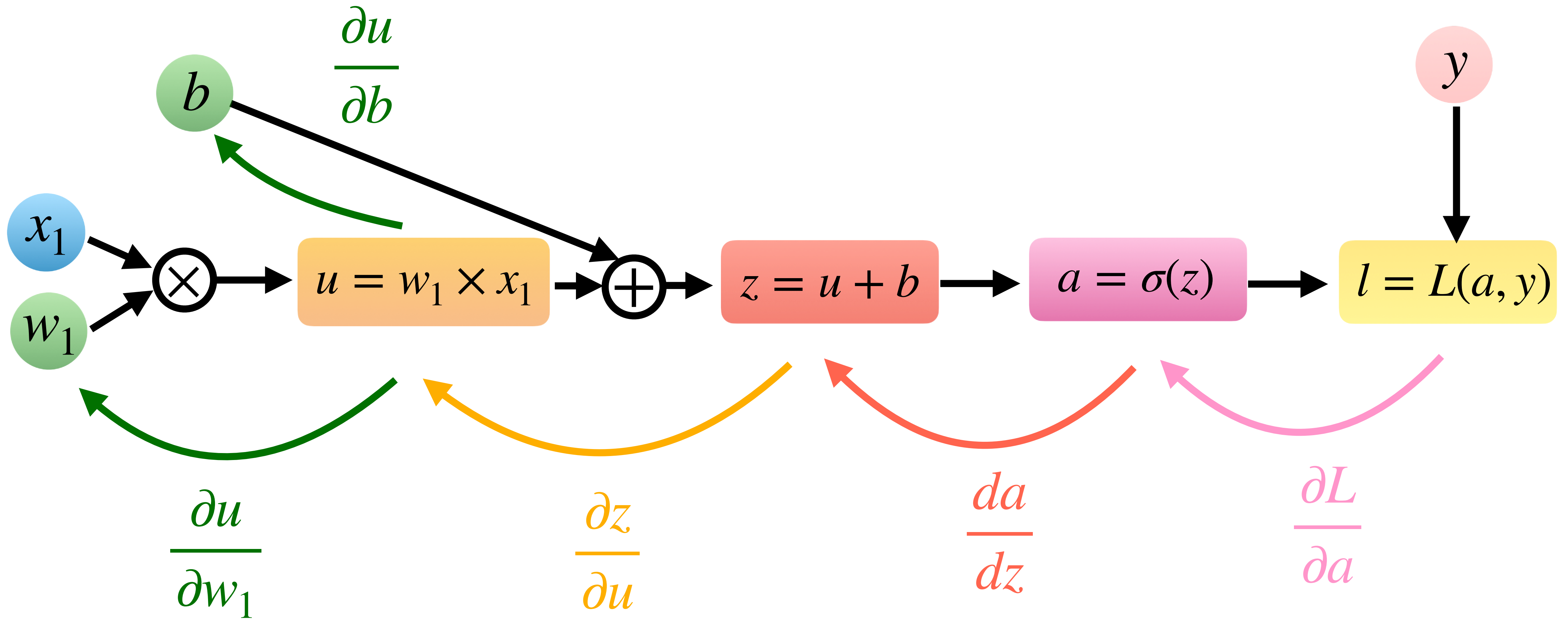
3.3

Model Training with Stochastic Gradient Descent

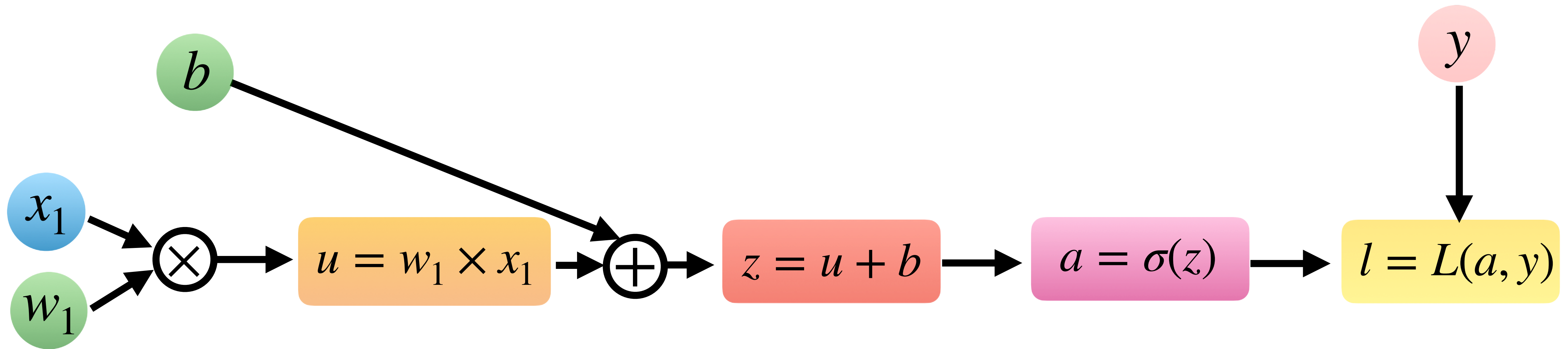
Part 2: Derivatives vs Gradients

Sebastian Raschka and the PyTorch Lightning Team

Chain Rule



Chain Rule



$$\frac{\partial L}{\partial w_1} = \frac{\partial u}{\partial w_1} \times \frac{\partial z}{\partial u} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

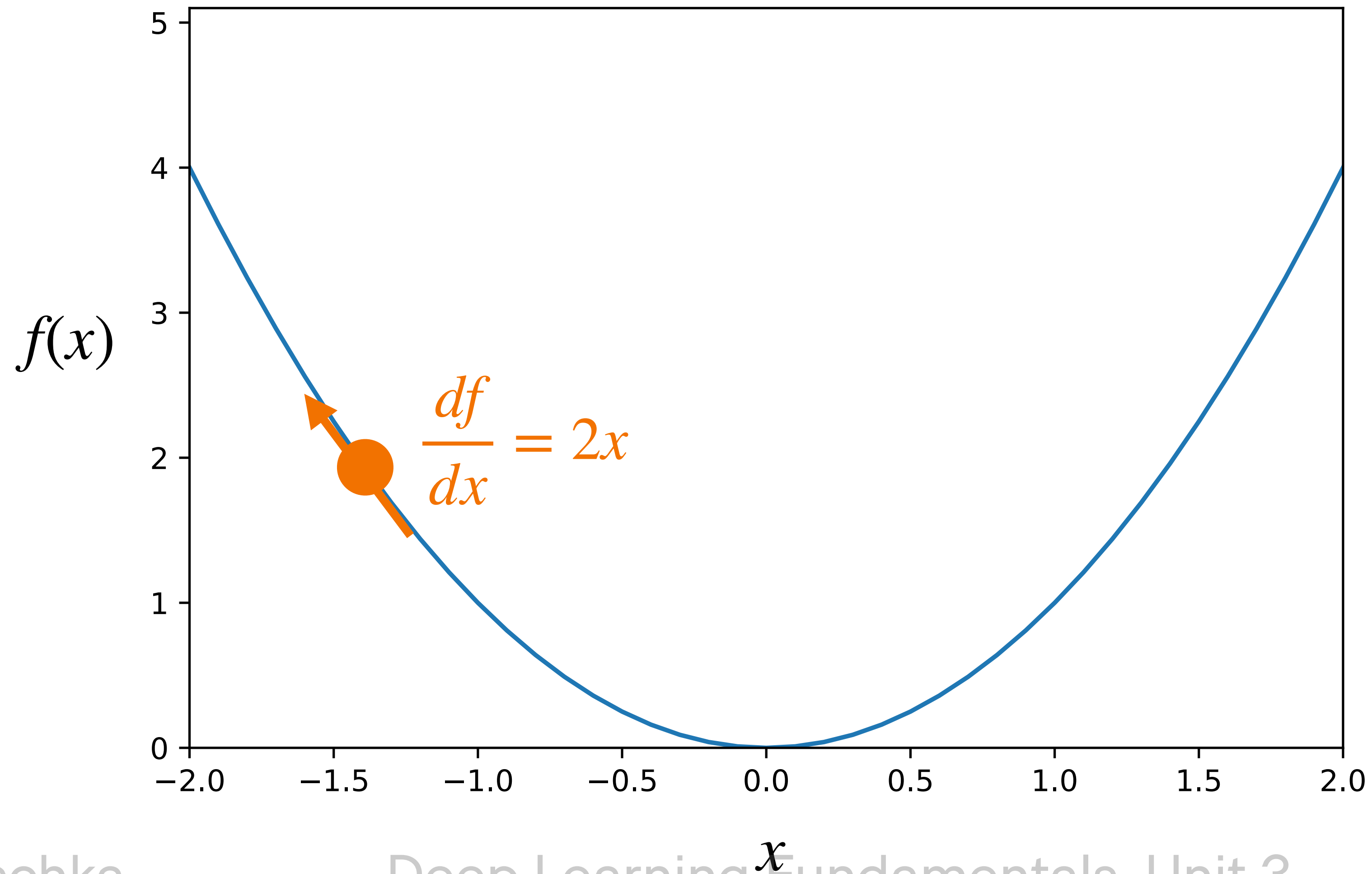
Derivatives vs Gradients

Derivative

$$f(x) = x^2$$

$$\frac{df}{dx} = 2x$$

$$f(x) = x^2$$



Derivatives vs Gradients

Derivative

$$f(x) = x^2$$

$$\frac{df}{dx} = 2x$$

Partial derivative

$$f(x, y) = x^2 + y$$

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\partial f}{\partial y} = 1$$

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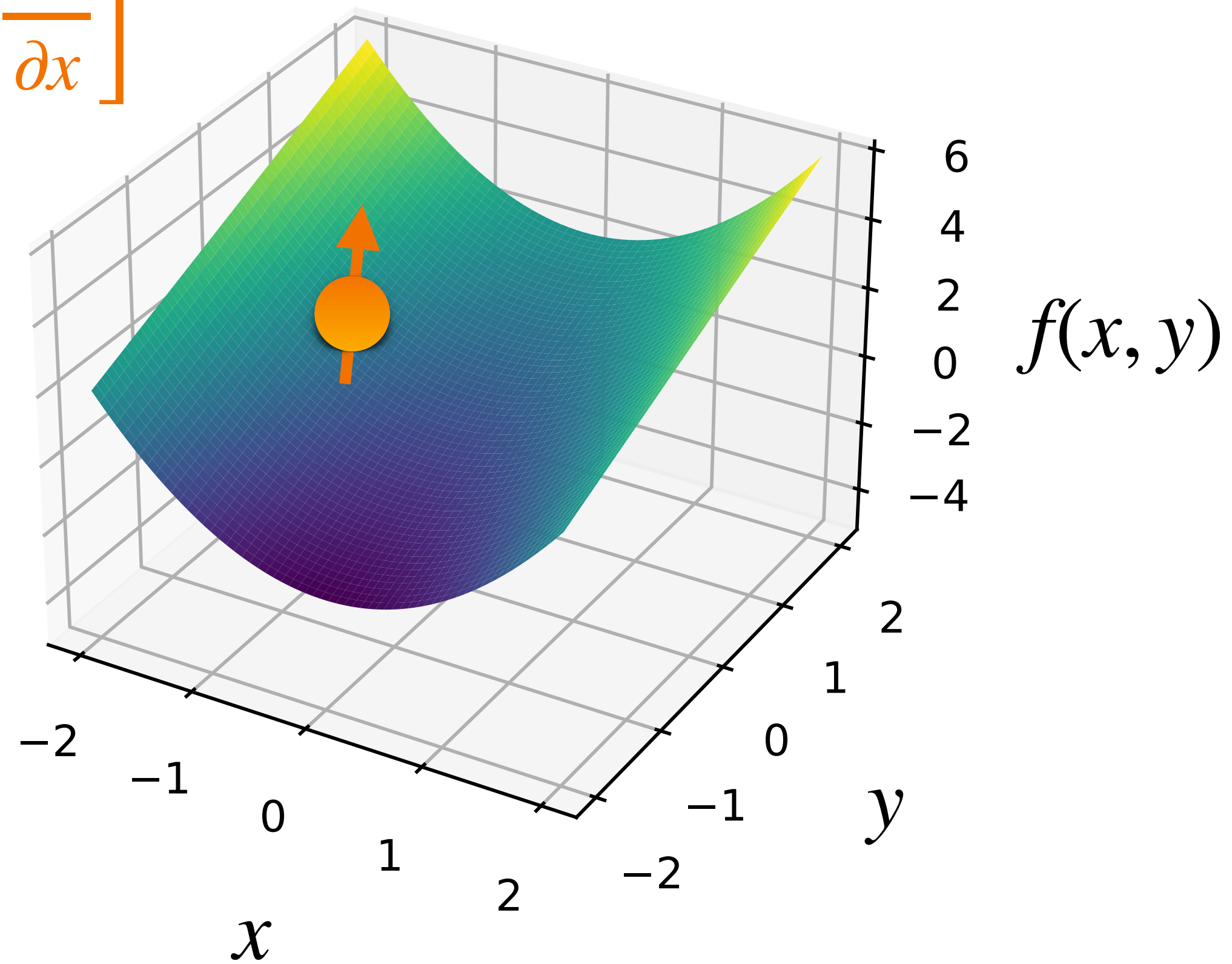
Gradient

$$f(x, y) = x^2 + y$$

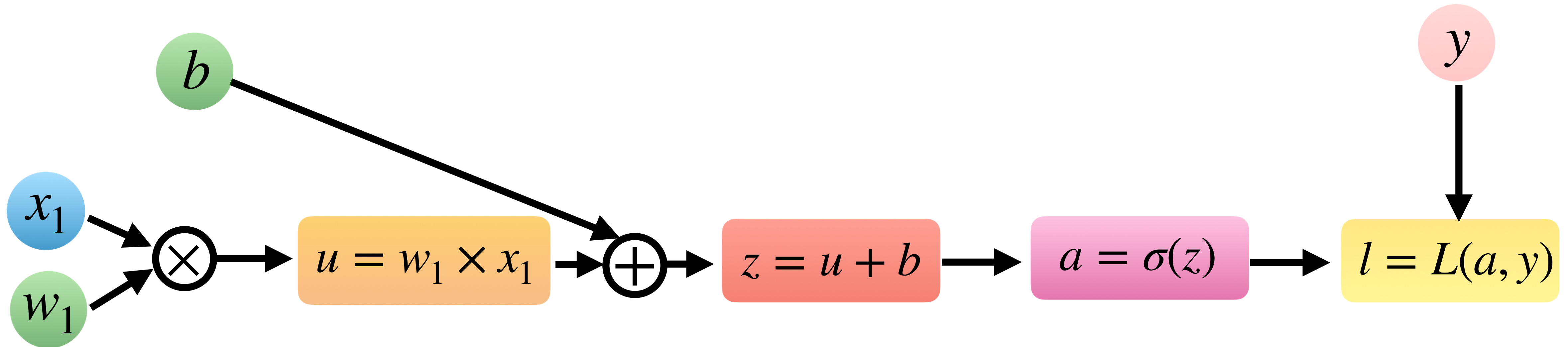
$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x \\ 1 \end{bmatrix}$$

$$f(x, y) = x^2 + y$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \end{bmatrix}$$

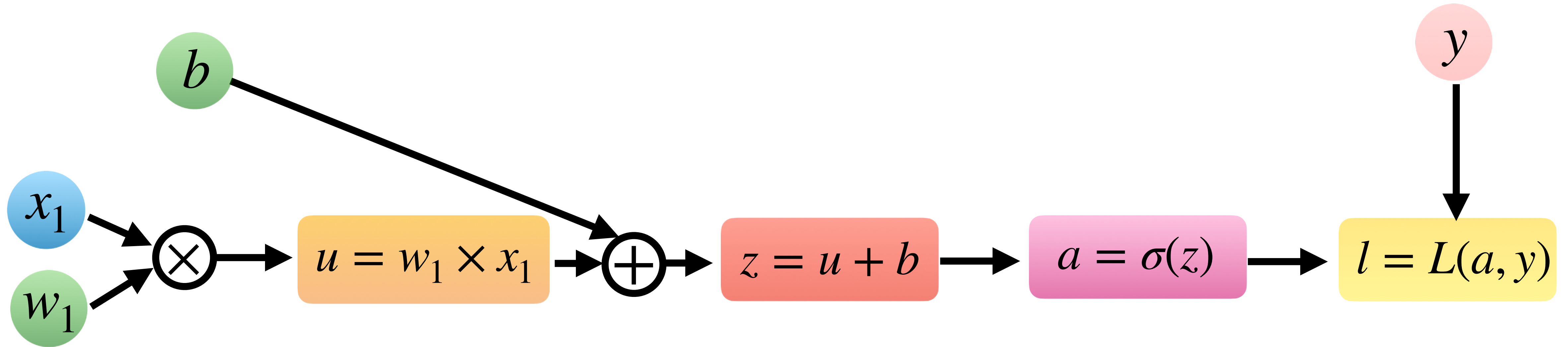


Chain Rule



$$\frac{\partial L}{\partial w_1} = \frac{\partial u}{\partial w_1} \times \frac{\partial z}{\partial u} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

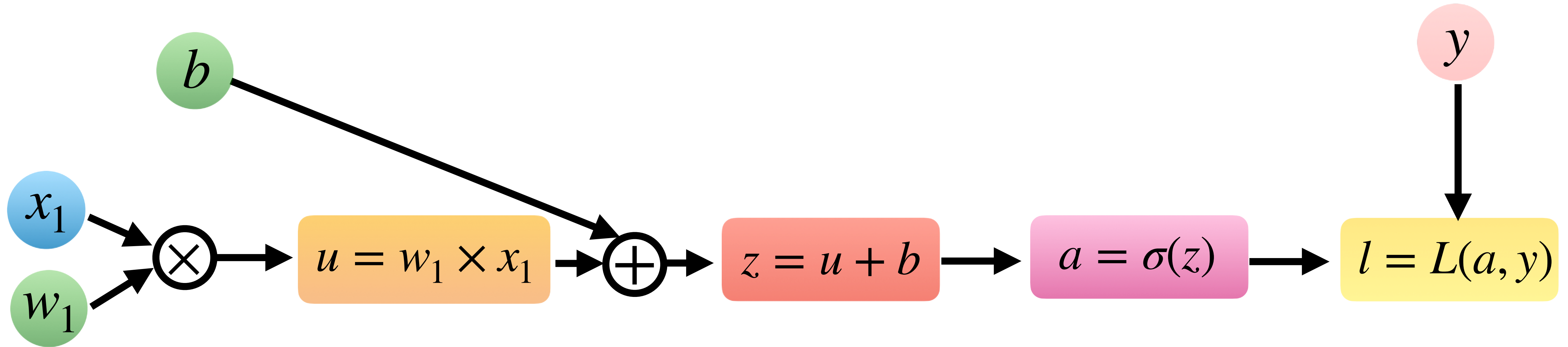
Chain Rule



$$\frac{\partial L}{\partial w_1} = \frac{\partial u}{\partial w_1} \times \frac{\partial z}{\partial u} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

$$\frac{\partial L}{\partial b} = \frac{\partial z}{\partial b} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

Chain Rule



$$\frac{\partial L}{\partial w_1} = \frac{\partial u}{\partial w_1} \times \frac{\partial z}{\partial u} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

$$\frac{\partial L}{\partial b} = \frac{\partial z}{\partial b} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial b} \end{bmatrix}$$

Minimizing the Loss Using Stochastic Gradient Descent